

Multimodal AI Systems

Vision-Language Models: CLIP, LLaVA, GPT-4V, Claude 3

Radford et al. CLIP (2021) | Liu et al. LLaVA (2023) | Anthropic Claude 3

Overview

Multimodal AI models process and generate multiple modalities: text, images, audio, video, code, and structured data. Claude 3's vision capabilities, GPT-4V, Gemini, and LLaVA represent the current generation. This document covers architectures, use cases, and APIs.

1. Multimodal Architectures

CLIP (OpenAI, 2021): Contrastive Language-Image Pre-training. Trains image encoder + text encoder jointly to maximize cosine similarity of matching pairs. Enables zero-shot classification and image-text retrieval. Foundation of most modern multimodal systems.

Flamingo (DeepMind, 2022): Visual Language Model. Freezes a pre-trained vision encoder (NFNet) and language model (Chinchilla), adds cross-attention layers (Perceiver Resampler) between them. Enables few-shot multimodal in-context learning.

LLaVA (Liu et al., 2023): Simple, effective open-source VLM. Vision encoder (CLIP ViT-L) → linear projection → LLM (LLaMA). Trained on GPT-4 generated image descriptions. Strong reasoning.

Claude 3 Vision (Anthropic, 2024): Natively multimodal. Accepts images alongside text. Trained end-to-end. 200K context window with up to 20 images. Excels at document analysis, charts, diagrams.

GPT-4o (OpenAI, 2024): Unified omnimodal model: text + image + audio in a single model. Real-time audio with <300ms latency. Any-to-any generation.

2. Claude Vision API

```
import anthropic, base64

client = anthropic.Anthropic()

# Encode image
with open("chart.png", "rb") as f:
    img_data = base64.standard_b64encode(f.read()).decode("utf-8")

response = client.messages.create(
```

```

model="claude-opus-4-5",
max_tokens=1024,
messages=[{
    "role": "user",
    "content": [
        {
            "type": "image",
            "source": {
                "type": "base64",
                "media_type": "image/png",
                "data": img_data,
            },
        },
        {
            "type": "text",
            "text": "Analyze this chart. What are the key trends? "
                    "Extract the exact values and provide recommendations."
        }
    ],
}]
)
print(response.content[0].text)

```

3. Use Cases for Multimodal AI

Document Intelligence: Extract text, tables, signatures from PDFs, invoices, contracts

Medical Imaging: Analyze X-rays, pathology slides (research context); FDA compliance required

Code Screenshot to Code: Convert UI mockups or app screenshots to working code

Chart/Data Analysis: Read charts, extract data points, generate analysis narratives

Visual RAG: Embed document pages as images; retrieve by visual similarity (ColPali)

Product Search: Find similar products from photo uploads (e-commerce)

Accessibility: Auto-generate alt-text for images; describe scenes for visually impaired